

Aadit Kannan

Selected Hardware Projects

Mechanical design, robotics hardware, fabrication, electronics integration, and test.

LINKS

aaditkannan.com

aaditkannan.com/projects

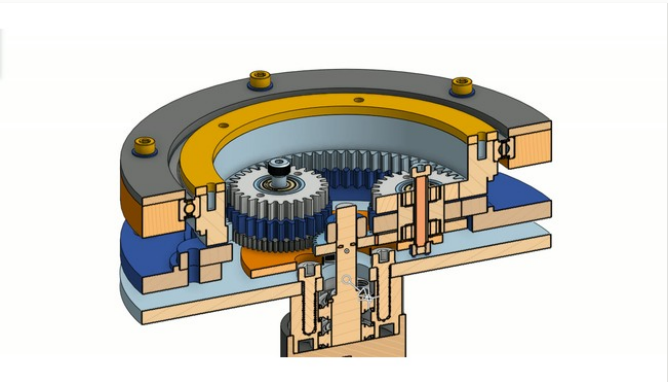
aaditkannan@berkeley.edu

[linkedin.com/in/aaditkannan](https://www.linkedin.com/in/aaditkannan)

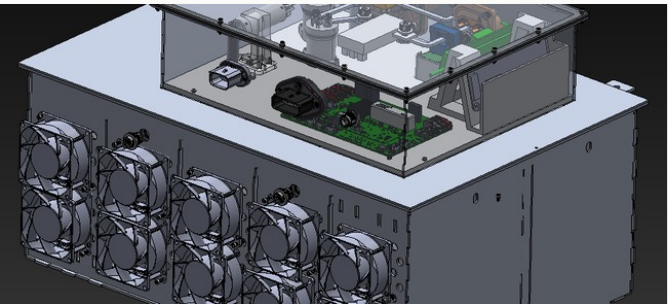
github.com/aaditkannan

PROJECT INDEX

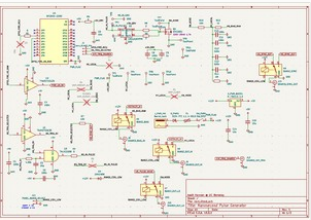
- 1 Wolfrom Compound Planetary Actuator
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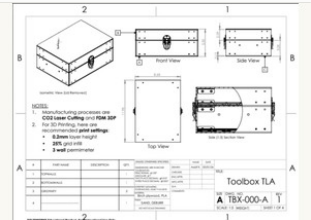
Wolfrom actuator section render



588V accumulator CAD



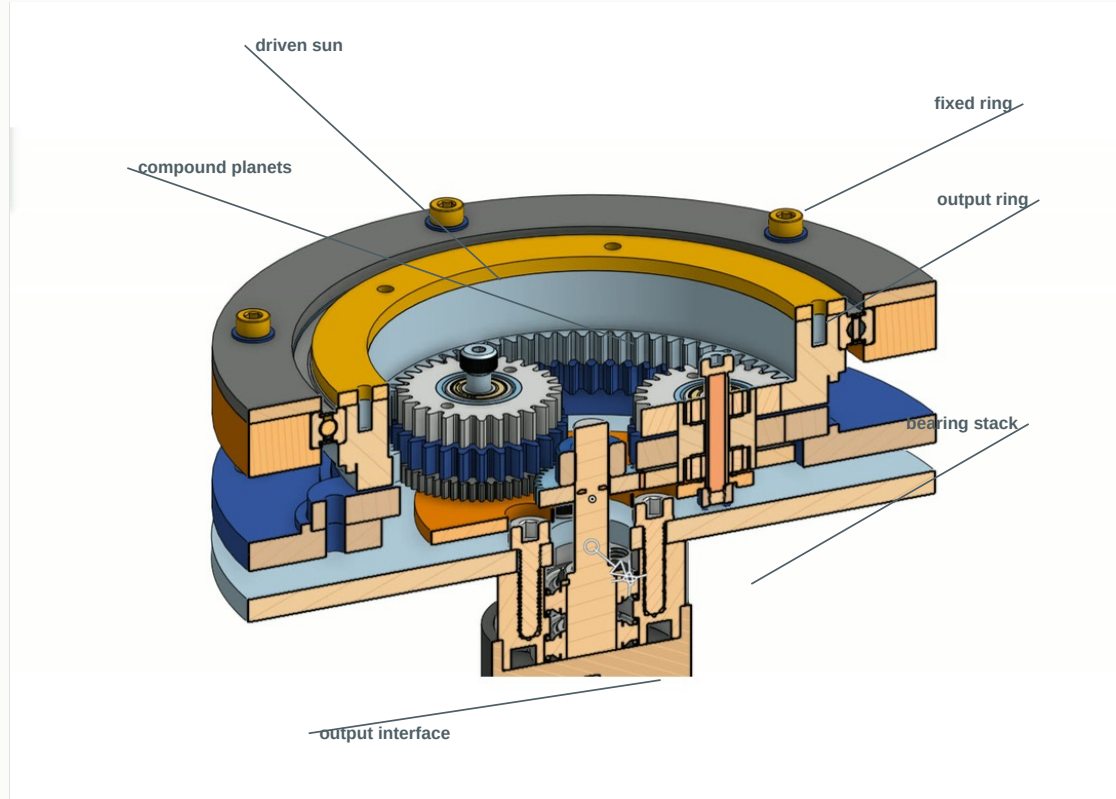
ns pulse generator schematic



Toolbox drawing package

Wolfrom Compound Planetary Actuator

Compact 50:1 compound Wolfrom gearbox for robotic joints. The motor drives a sun gear; each compound planet meshes with a fixed internal ring on one side and an output internal ring on the other. The small tooth-count difference at the ring meshes creates high reduction in one compact axial package.



CAD section showing the sun, compound planets, fixed ring, output ring, bearings, and output interface.

MY ROLE

Designed the gearbox architecture, actuator stackup, bearing support, output ring split, fastener access, and prototype assembly sequence.

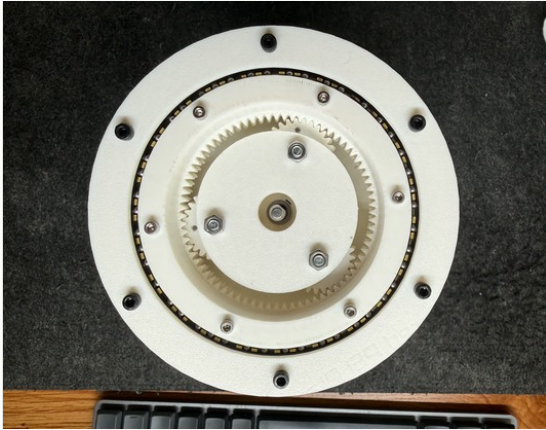
ARCHITECTURE DECISIONS

- Used a compound planetary Wolfrom layout instead of stacking multiple conventional stages.
- Kept the actuator axial package compact around the sun, carrier, rings, bearings, and output hub.
- Split the output ring and designed around bolt access so the prototype can be assembled and re-timed.
- Designed the bearing stack and output interface around printed prototypes first, then a metal revision path.

WHY IT MATTERS FOR ROBOTIC JOINTS

The relevant tradeoff is packaging high reduction while keeping friction, reflected inertia, assembly tolerance, and output support under control. The project is still in prototype iteration; the value is the architecture and mechanical packaging work, not a finished actuator claim.

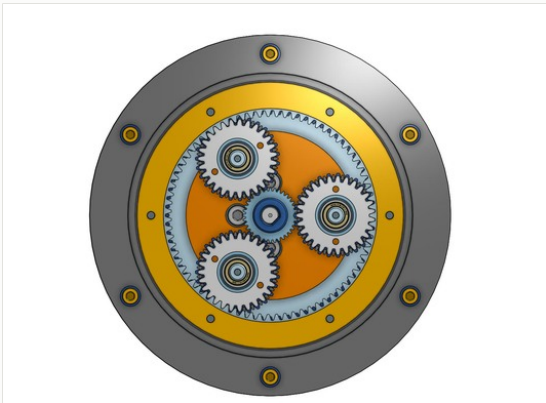
Wolfrom Actuator - Prototype Status



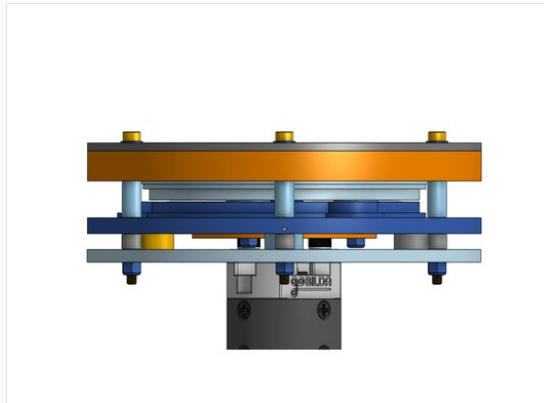
Printed prototype used for gear mesh, stack height, and assembly checks.



Bench assembly work: checking fit-up and hand assembly sequence.



Top view of compound planets, output ring, and fastener pattern.



Side view of the plate stack, motor interface, and axial package.

CURRENT DESIGN DECISIONS

- Rebalanced the stage split so less reduction happens in the high-loss ring differential, reducing internal power recirculation compared with a naive Wolfrom split.
- Targeting about 50:1 reduction; current tooth counts produce about 50.45:1.
- Working toward near-zero backlash through split-gear preloading rather than claiming final measured backlash.
- Printed prototypes are being used for gear mesh, packaging, stack height, and assembly checks.

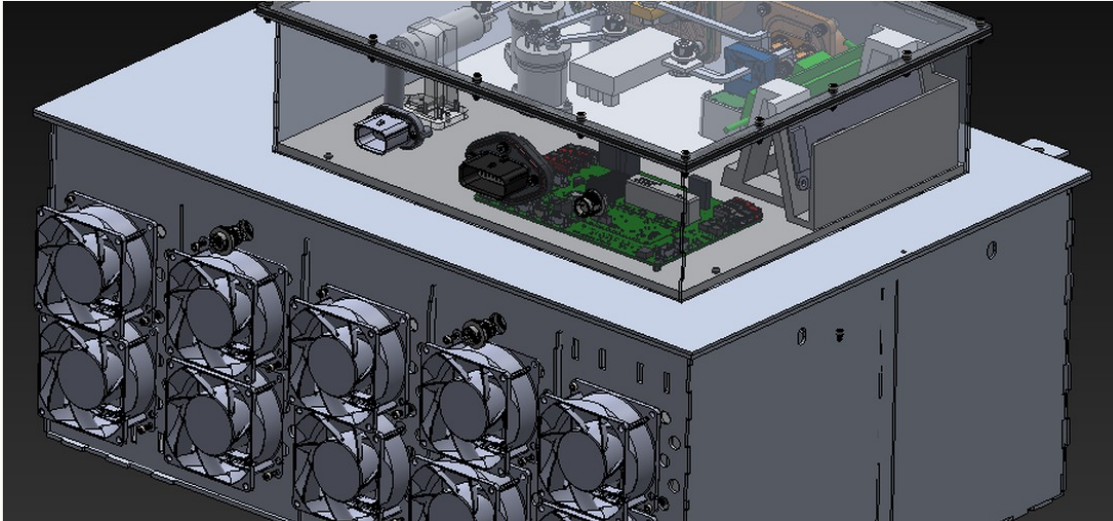
IN PROGRESS

- KISSsoft refinement for macro-geometry, profile shifts, and tip relief on high-load ring meshes.
- Next mechanical revision: CNC aluminum structure, wire-EDM rings, hardened pins, and needle bearings.
- Planned dyno characterization for torque, efficiency, and backdrive behavior after the metal version is built.

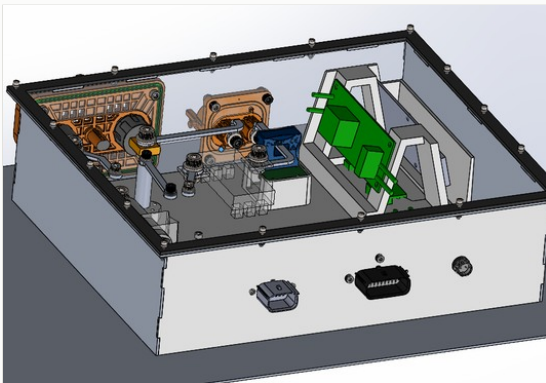
STATUS

Printed prototype built and fit-checked; metal revision, KISSsoft refinement, and dyno characterization are in progress.

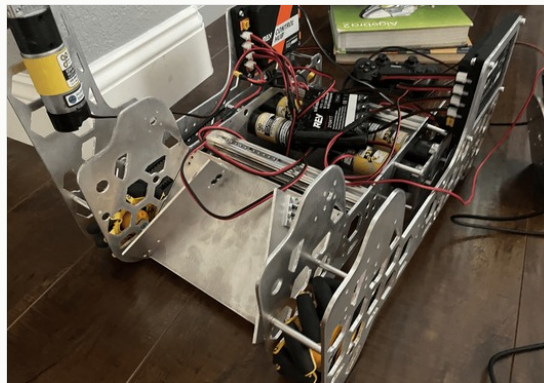
Formula Electric 588V Accumulator Hardware



Accumulator CAD showing HV electronics attic, fans, service access, and enclosure packaging.



Electronics attic layout and board/connector packaging.



Physical hardware packaging context for accumulator integration.

MY ROLE

Designed and manufactured the HV electronics attic enclosure and busbar packaging for a 588V accumulator, focusing on insulation, sealing, access, mass, and review documentation.

ENGINEERING WORK

- Packaged the HV electronics attic, busbars, boards, fans, fasteners, and service interfaces inside the accumulator envelope.
- Preserved insulation clearances, cooling paths, board access, fastener access, and inspection access while reducing attic mass by 18%.
- Designed around waterproofing and high-dielectric insulation requirements for a 588V battery system.
- Routed busbars for 80A peak discharge and prepared design material for SES/ESF review.

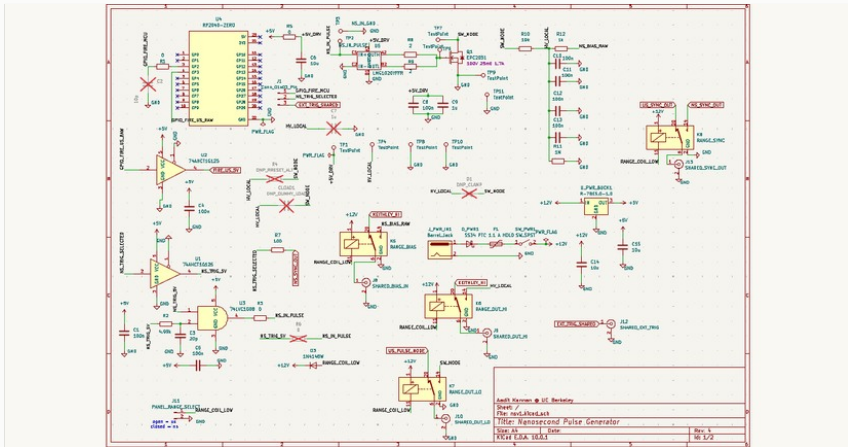
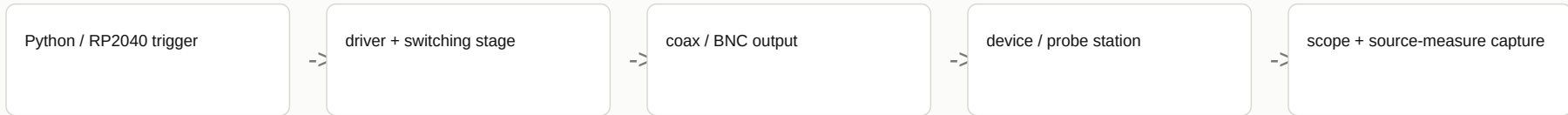
DESIGN CONSTRAINTS

The packaging problem was not just fitting electronics in a box: it had to leave room for service, cooling, inspection, sealing, high-voltage isolation, and manufacturable fastener access.

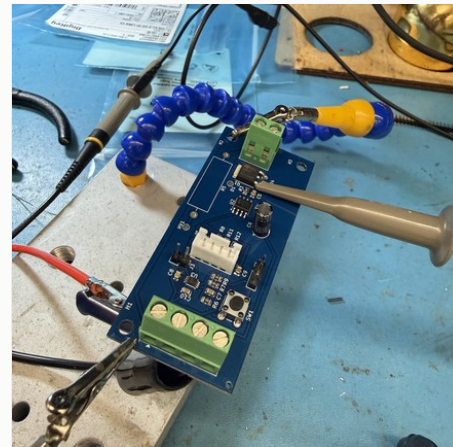
Ramesh Lab ns/us Pulse Generator PCB

Instrument for ferroelectric and RF characterization of BiFeO₃ thin-film devices. The goal is controlled nanosecond and microsecond pulse generation with grounded lab I/O, source-measure biasing, RP2040/Python triggering, and oscilloscope validation.

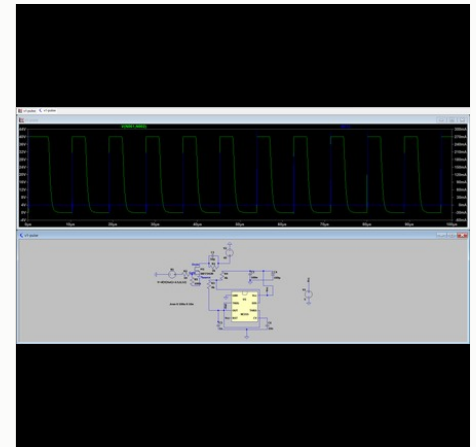
SYSTEM FLOW



Current nanosecond pulse-generator schematic.



V1 microsecond board during bench probing.

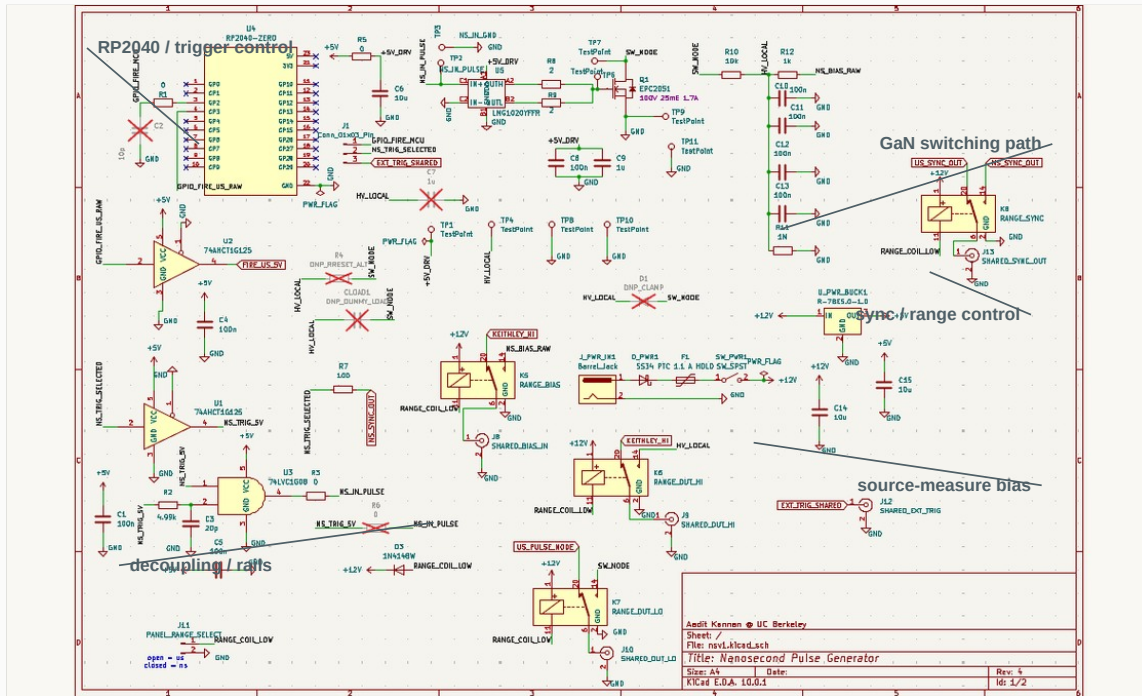


V1 timing/switching simulation used during early design work.

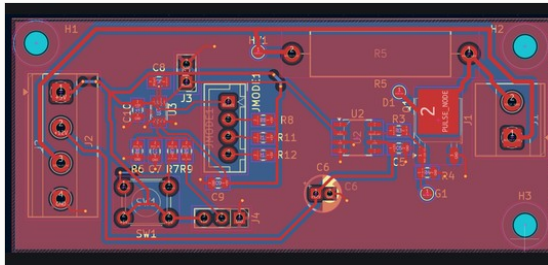
DESIGN FOCUS

- Separate ns and us pulse paths around the needs of thin-film switching and transport tests.
- Use GaN switching and clean trigger/sync routing instead of loose bench wiring.
- Expose grounded coax/BNC-style I/O and source-measure biasing for probe-station work.

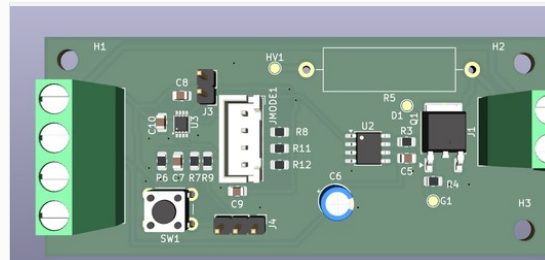
Pulse Generator - Schematic, Layout, Test



KiCad schematic for the current ns/us pulse generator revision.



V1 PCB layout used to check routing, connector placement, and return paths.



V1 rendered board: useful as a physical integration reference, not the current ns revision.

V1 LEARNING ARTIFACT

The first microsecond board validated the basic bench workflow, but exposed integration issues: clip-lead power, weak lab I/O, and awkward triggering.

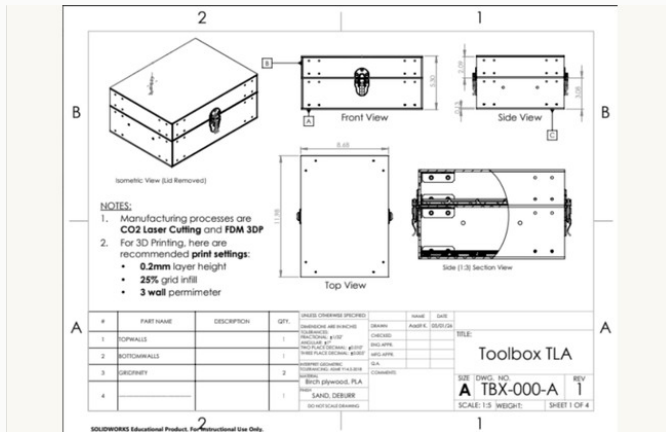
CURRENT REVISION

- Folds nanosecond and microsecond pulse generation into a cleaner instrument.
- Adds coax output, source-measure biasing, shared trigger/sync paths, and Python/RP2040 control.
- Bring-up plan: verify rails and trigger logic, validate switching into dummy loads, then move toward device fixtures.
- No final rise-time, jitter, amplitude stability, or device-result claims yet.

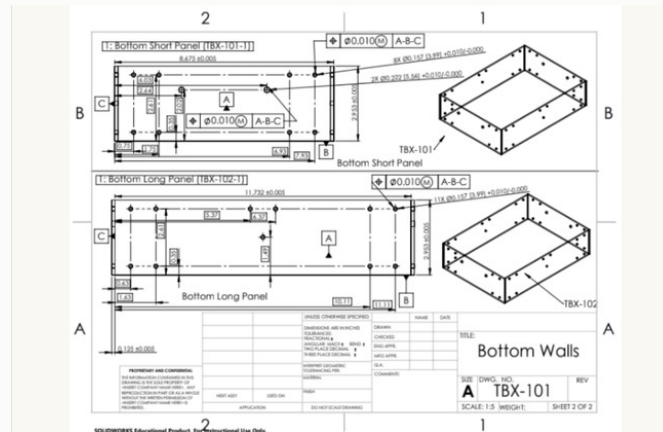
TOOLS

KiCad, LTspice, RP2040, Python, LabVIEW, JupyterLab, oscilloscope, source-measure equipment, coax/BNC lab interfaces.

Custom Toolbox Design & Manufacturing



Assembly drawing defining lid, latch, hinge, body, and insert interfaces.



Part drawing with datum references and positional tolerance callouts.

MY ROLE

Used the toolbox as a controlled manufacturing exercise: model the assembly, define functional datums, create buildable drawings, fabricate parts, and inspect fit-up instead of relying on hand-fitting.

ENGINEERING WORK

- Created production-style drawings using ASME Y14.5-2018 GD&T conventions.
- Specified datums, position tolerances, fastener clearances, and lid/hinge/latch interfaces.
- Designed around laser-cut plywood panels, FDM inserts, and fastened joints visible in the prototype.
- Checked fit-up using calipers, fasteners/gauge pins, squareness, lid closure, and hinge/latch alignment.

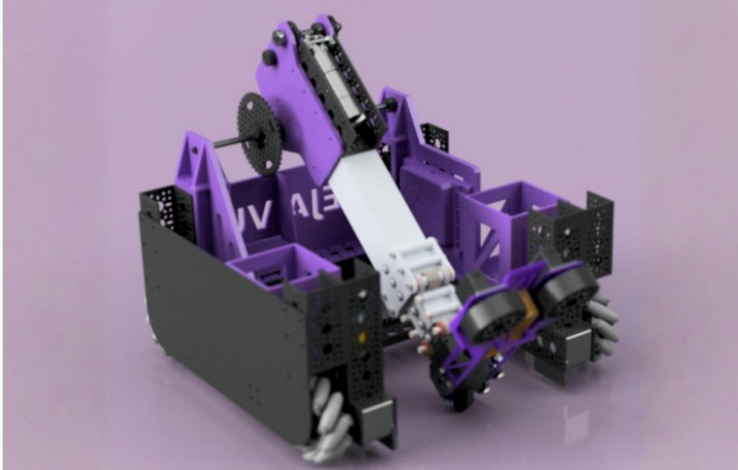


Prototype fit-up after fabrication with hinges, latches, handle, and fasteners.

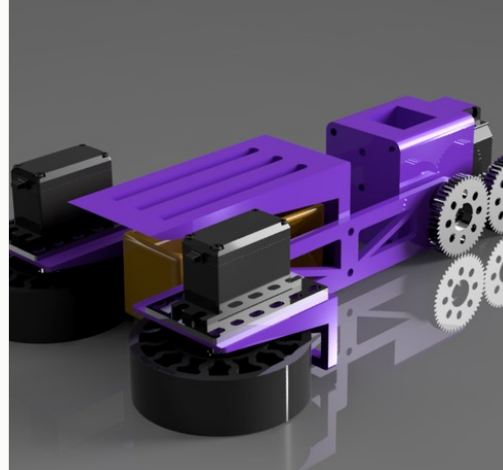


Open prototype showing interior fit-up; final-version photos unavailable.

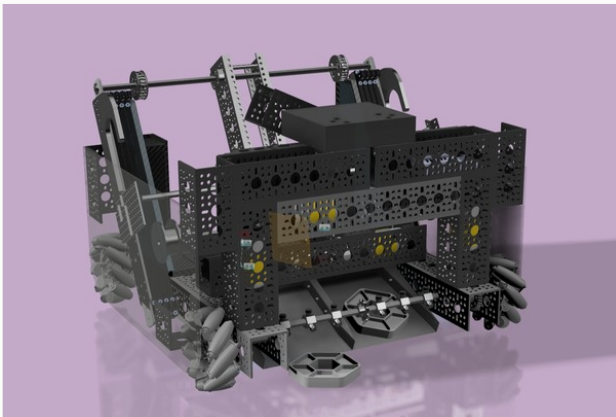
FIRST Robotics Hardware



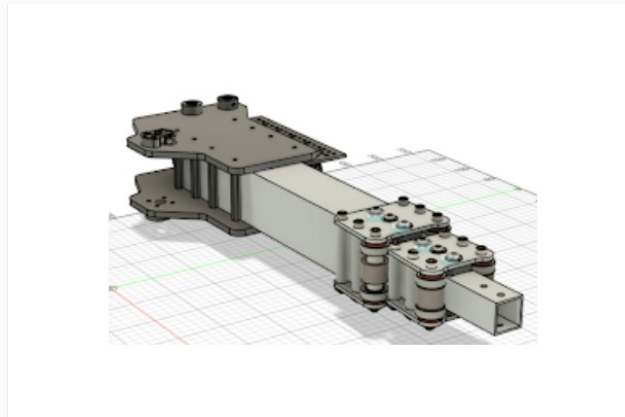
INTO THE DEEP robot CAD and subsystem packaging.



High-iteration intake mechanism CAD.



CENTERSTAGE robot CAD assembly.



Linkage and end-effector packaging detail.

MY ROLE

Hardware Lead for FTC #13216 and captain/founder work across FIRST teams. Focus here is mechanism design, reliability, serviceability, and iteration under match constraints.

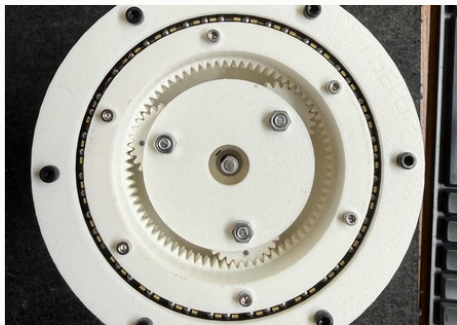
ENGINEERING WORK

- Designed 8 robots and 500+ part CAD assemblies across drivetrain, intake, transfer, arm/lift, end effector, and hang systems.
- Used FEA to guide custom aluminum and printed components for stress and weight.
- Built with printed parts, laser-cut plates, CNC-machined aluminum, and shop-fabricated hardware.
- Wrote Java TeleOp for field-centric mecanum drive, PID loops, mechanism sequencing, and driver feedback.
- Won Robot Design Award 3 times; mentored 30+ members in CAD and Java.

Tools + Methods

CAD / Mechanical

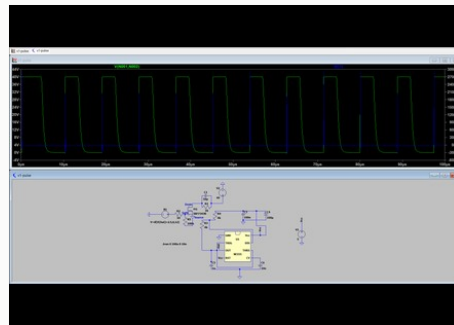
- SolidWorks
- Onshape
- GD&T
- DFM/DFA
- FEA
- mechanism packaging



Actuator stackup prototype

Manufacturing

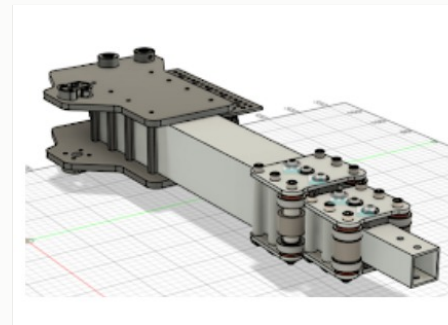
- mill
- lathe
- waterjet
- laser cutter
- TIG
- FDM/SLA
- fastener fit-up



Pulse simulation workflow

Electronics / Test

- KiCad
- LTspice
- oscilloscope
- source-measure
- RP2040/Python
- coax/BNC interfaces



Mechanism packaging

Analysis / Software

- KISSsoft
- MATLAB/Simulink
- Python
- JupyterLab
- LabVIEW
- data logging

LINKS

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